

Practice Questions for 05L2-ML Basics

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1 Questions

1. According to the lecture, what is the core idea behind machine learning?
 - A. Solving complex mathematical equations
 - B. Automated inductive reasoning
 - C. Creating complex algorithms
 - D. Simulating human thought processes
 - E. Developing new programming languages
2. In the context of linear regression, what does the term "loss" refer to?
 - A. The number of data points used for training
 - B. The difference between predicted and actual values
 - C. The complexity of the model
 - D. The time it takes to train the model
 - E. The number of features used in the model
3. What is the main purpose of the Perceptron algorithm?
 - A. To perform linear regression
 - B. To perform clustering
 - C. To find a separating hyperplane for linearly separable data
 - D. To predict continuous values
 - E. To reduce the dimensionality of data

2 Answers

1. According to the lecture, what is the core idea behind machine learning?
 - A. Answer A is incorrect because while math is used, it is not the core idea.
 - B. Answer B is correct because the lecture defines machine learning as automated inductive reasoning.
 - C. Answer C is incorrect because while algorithms are used, it is not the core idea.
 - D. Answer D is incorrect because while it is related, it is not the core idea.
 - E. Answer E is incorrect because while programming is used, it is not the core idea.
2. In the context of linear regression, what does the term "loss" refer to?
 - A. Answer A is incorrect because the number of data points is not the loss.
 - B. Answer B is correct because the loss function measures the difference between the predicted and actual values.
 - C. Answer C is incorrect because the complexity of the model is not the loss.
 - D. Answer D is incorrect because the training time is not the loss.
 - E. Answer E is incorrect because the number of features is not the loss.
3. What is the main purpose of the Perceptron algorithm?
 - A. Answer A is incorrect because the Perceptron algorithm is not for regression.
 - B. Answer B is incorrect because the Perceptron algorithm is not for clustering.
 - C. Answer C is correct because the Perceptron algorithm is designed to find a separating hyperplane for linearly separable data.
 - D. Answer D is incorrect because the Perceptron algorithm is not for predicting continuous values.
 - E. Answer E is incorrect because the Perceptron algorithm is not for dimensionality reduction.